

AWARE: Approaches in WetlAnd Restoration - focus on fen landscapes • Warsaw, Poland 21-23 April 2013

RESTORATION OF A MEDITERRANEAN DRAINED PEATLAND: THE CASE STUDY OF THE MASSACIUCCOLI LAKE BASIN (TUSCANY, IT)

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Abstract: The Massaciuccoli lake basin (Tuscany, Italy) is characterized by a Mediterranean climate. A large part of this basin is an artificially drained coastal floodplain affected by large nutrient loading coming from agricultural activities. Due to complex hydrological setting and high soil organic matter content (up to 50%), problems have arisen, such as phosphorus leaching and land subsidence. In the project that started in 2011, rewetting of a part of the area was tested as a solution for restoring lost ecological functions of this site. In a pilot experimental field of 15 ha, three different management systems, with an increasing anthropogenic impact has been tested: natural wetland, constructed wetland and vegetation filters. In addition, a conventionally cultivated and an uncultivated drained peat soil, characterized by a natural vegetation succession , were used as controls. Hydrological cycle, surface- and groundwater quality, peat oxidation rate, microbial diversity and functionality, CH_4 and CO^2 emissions, plant nutrient removal, biomass production and energy efficiency have been monitored in order to assess the most effective and sustainable management system.

Keywords: Mediterranean peatland, eutrophication, identifying appropriate conservation and restoration objectives, nutrient removal, increasing public support and participation

Introduction

The Massaciuccoli Lake floodplain is located in the Natural Park of San Rossore, Migliarino and Massaciuccoli (Figure 1), which is one of the most important residual coastal marshy areas of the Tuscany (Italy). Since the 1930s, a large part of the Massaciuccoli floodplain has been drained for agricultural purposes. To ensure a water table depth suitable for cultivation, a complex network of artificial drains and pumping stations has been used to drain the superficial aquifer and rainwater. In the drained areas, cultivated peat soils (autri-sapric and endo-salic histosoils), with values of organic matter reaching up to 50% in some cases, are present (Pistocchi et al. 2012). Land use is characterised by conventional agriculture (covers 80% of the area) and periurban infrastructures, such as a wastewater treatment plant. In the peatland area, cropping systems are based on continuous production of maize (*Zea mays* L.), sunflower (*Heliantus annuus* L.), wheat (*Triticum spp*. L.) or maize-wheat rotations, while winter cereals are mainly cultivated in the remaining part of the basin (Silvestri et al. 2012). As a consequence of land use, several environmental concerns arose in the last 50 years. The most important concerns are those related to:

I. eutrophication of the lake due to nutrient enrichment (N, P) in the surface- and groundwater. Indeed, from the 1970s, the lake, from an initial oligotrophic status, progressively turned into an eutrophic/hypereutrophic system;

II. the subsidence rate (2-3 m in 70 years) due to compaction and increased mineralization of peat. This process, started since land reclamation, left the lake perched above the drained area, which is now 0 to 4 m below the sea level (Rossetto et al. 2010).

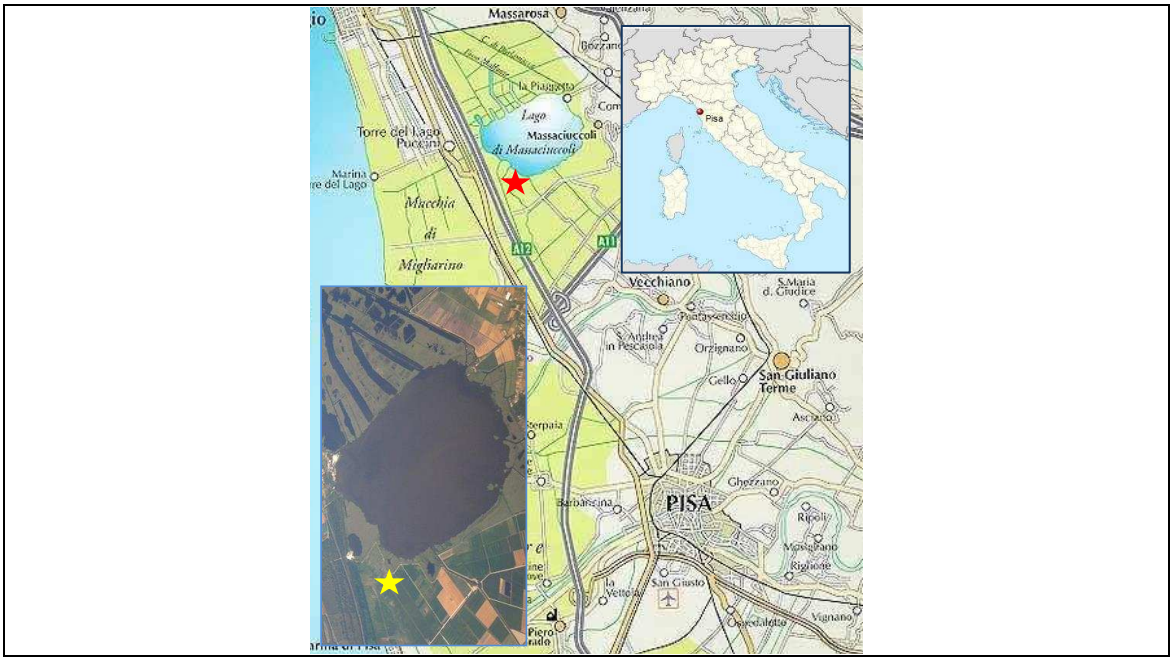


Figure 1. Localisation of the area of the Massaciuccoli Lake basin (red star) and focus on the pilot experimental field (yellow star).

Materials and methods

The project RestoMedPeatland (https://sites.google.com/site/restomedpeatland/) that was started in 2011, identified rewetting and setting-up a phyto-treatment system as the solution for improving water quality, and restricting soil organic matter (SOM) mineralisation, and, therefore, a method to restore the ecological functions of this site. A pilot experimental field of 15 ha was set-up and three different management systems, with increasing anthropogenic impact, has been tested (Figure 2a,b): constructed wetland (A), vegetation filters (B) and natural wetland (C). This implies a gradient in regulation of water regime (from a strongly controlled system to a "quasi" natural rewetting), plant communities (from cultivated to native communities) and harvesting strategy. The soil-plant continuum systems are expected to reduce nutrient load. In addition, a conventionally cultivated (D) and an uncultivated drained (E) peat soil (the latter characterized by a natural vegetation succession), were used as controls.

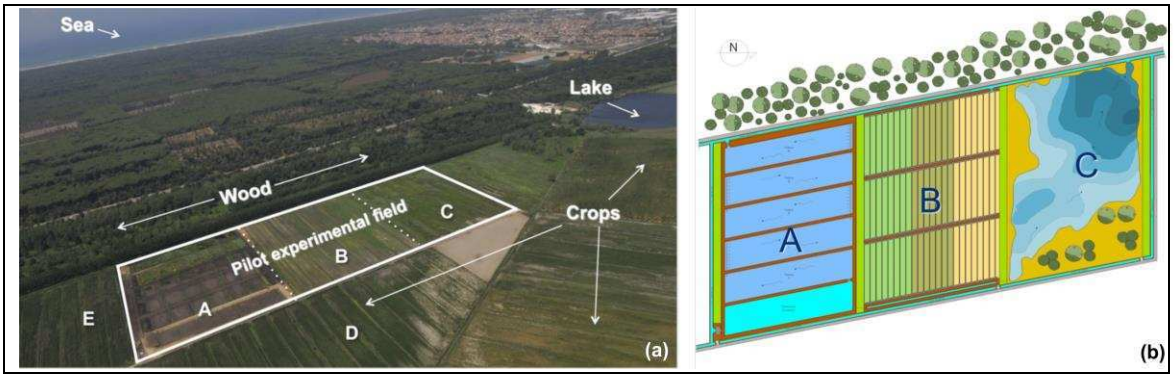


Figure 2. Areal view (a) and set-up (b) of the pilot experiment with three different management systems: constructed wetland (A), vegetation filters (B) and natural wetland (C). The conventionally cultivated (D) and the uncultivated drained peat soil (E) were used as controls are shown in (a).

System A was internally and externally banked (0.4-0.5 m) in order to force a convoluted pattern of water flow, which results in a lengthening of travel time. The entrance flow was controlled to achieve the targeted water quality. Water management may require different flow regimes depending on the variability of the factors controlling the outlet discharge, such as rainfall, infiltration or evapotranspiration. *Phragmites australis* L. and *Thypha angustifolia* L. were the main plants in the system and the management consists of harvesting the biomass according to the system functionality. System B was based on the plantation of seven different no-food crop plants, managed by periodic cutting and biomass harvesting. The system is not dammed, but crossed by a dense network of small channels. These supply the drainage water to the crops through lateral infiltration. A three types of plants were used : i) woody plants: *Populus spp*., *Salix spp*., ii) perennial grasses: *Arundo donax* L., *Miscanthus x giganteus* Greef et Deuter, *P. australis* L.; iii) turfgrasses: meadows of cool-season grasses (*Festuca arundinacea* L., *Poa pratensis* L., *Lolium perenne* L.) and seashore paspalum (*Paspalum vaginatum* Swartz). System C was set-up as a rewetted area, where spontaneous re-colonization of the vegetation takes place. Natural elevation variation helped to create areas of different slope in order to promote colonization by different plant species. In order to increase the efficiency of the phyto-treatment system floating structures were introduced, on which plant species with spread roots can be planted, that have a high water purification potential.
In order to assess the effectiveness of the different phyto-management systems, two control areas were identified, next to the experimental plots. These areas represented a cultivated (D) and an uncultivated (E) drained peat soil with a natural vegetation succession. To assess the effectiveness of management system a multidisciplinary approach has been used, aiming to evaluate the status and changes of water, soil, belowground microbes, CH_4 and CO^2 soil emissions, plant communities and fauna biodiversity.

● Water: A complete hydrological monitoring has been performed starting with acquisition of meteorological data and measuring inflow and outflow for each phyto-system, as well as exploring the relationships between the superficial and deep aquifer (information on infiltration and/or exfiltration processes) and evapotranspiration. These data will be used in aunsaturated/saturated zone hydrological model to derive spatially-distributed data. As the system is a coastal one, the presence and influence of saline water on functionalities will be evaluated. Three sampling schemes are planned: 1) continuous monitoring with probes placed in situ; 2) a composite sampling with automatic samplers proportional to the flow pattern; 3) an instant discrete sampling for groundwater. The monitored parameters will be:

• Continuous: pH, oxygen, temperature, electrical conductivity (EC);

• Composite: EC, total suspended solids (TSS), total phosphorus (TP), total nitrogen (TN, Kjeldahl), nitrates (NO^{3-}), ammonium (NH^{4+}), dissolved organic carbon (DOC), dissolved inorganic carbon (DIC), anions, cations.

● Soil: A set of the soil physico-chemical properties (e.g. pH; EC; TN and available nitrogen (Navail); NO^{3-} and $NH^{4+;}$ TP, available and organic phosphorus (Pavail and Porg); C:N ratio; SOM and soil texture) has been measured in order to assess the soil status before rewetting. Measurements will be repeated after setting-up phyto-treatment system, in order to evaluate the impact of the different management on soil physico-chemical quality.

● Belowground microbes: Molecular methods based on polymerase chain reaction (PCR) of DNA have been used to assess the genetic diversity of soil microbial communities (Pellegrino et al. 2012). First, the structure and composition of whole soil microbial community of the experimental area has been assessed using DNA sequencing and high-throughput molecular tools. The microbial monitoring was based on the phylotaxonomic diversity of the whole bacterial and fungal communities, of the bacterial and archaeal communities involved in the N cycle and intraradical arbuscular mycorrhizal fungi (AMF). Data were obtained before the experiment, in order to have a baseline of the soil microbial community composition and structure. These data will be used for evaluating the shifts in microbial community due to the different phyto-management systems. Furthermore, quantitative real time (qPCR) assays will be performed in order to measure the functional microbial diversity by quantifying the activity of

functional genes involved in the N and CH_4 cycles. After rewetting methanogenesis and CH_4-oxidation will be studied by exploring methanotrophic bacteria and methanogenic archaea and correlating that to the CH_4 emissions.

● CH_4 and CO^2 soil emissions: The main greenhouse gases (GHG) fluxes has been measured in situ, in order to: (1) assess the effect of rewetting and phyto-treatment setting-up on soil GHG emissions, taking into account the increase of CH_4 and the decrease of CO^2 production, due to a shift from aerobic to an anaerobic conditions; (2) improve the estimate of the SOM mineralization, in order to better understand the factors influencing subsidence and nutrient release in the peatland environment. A soil respiration partitioning experiment has been set-up, aiming to separate the heterotrophic component from the total soil respiration. To measure the fluxes, a manual chamber with an infrared gas analyser and a methane detector with a floating accumulation chamber have been used respectively for CO^2 and CH_4.

● Plant species communities: The different plant species have been monitored and harvested for recording biomass production, shoot N and P uptakes and evaluating the technological parameters for energetic purposes (e.g. ash, calorific value of biomass). This kind of monitoring will be especially important in system B, in which each crop will be harvested on the basis of its specific morpho-physiological characteristics. The results will be used to choose the most suitable crop for phyto-treatment and energetic purposes. The spontaneous vegetation has been harvested both in spring and late summer to evaluate its contribution to water quality.

● Fauna biodiversity: The effect of the three management system has been evaluated also in terms of the biodiversity of the wetland fauna, such as birds and amphibians.

Conclusions

Here, we presented the RestoMedPeatland project which invastigated three different management systems aiming to improve water quality and to reduce the subsidence of the Massaciuccoli Lake basin Mediterranean peatland. The effectiveness of the systems will be evaluated by comparing their performances with those of the control areas. Successful results and social acceptance will contribute to further increase of rewetted areas in the Massaciuccoli floodplain and other similar Mediterranean peatlands.

Acknowledgements

Authors thank the staff of the project sponsor ”Consorzio di Bonifica Versilia-Massaciuccoli” for the technical support in setting-up and management of the field experiment.

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